Table 3.1: Difference between dyes and stains.

Dyes	Stains
Dyes are a colouring agents used for general purposes.	Stains are colouring agents used for biological purposes.
Dyes are the textile colouring agents that are prepared with lesser specification and they may contain impurities.	Stains are pure. They are prepared with greater care and specification.

3.3.1 Classification of Stains

1. On the basis of origin, stains can be classified as natural and synthetic.

(i) Natural stains:

- These stains are obtained directly from natural products. For example, Haematoxylin is obtained from the heartwood of a tree (*Haematoxylon campechianum)*.
- The natural stains are used mainly for histological purposes.

(ii) Synthetic stains:

- These are artificially produced mainly from coal tar products and hence popularly called coal-tar dyes.
- A majority of stains used in microbiology are the synthetic type and manufactured from Aniline. For example, Crystal violet, Safranin, Methylene blue and Acid fuchsin.

2. On the basis of chemical behavior, dyes are classified as acidic, basic and neutral.

- An acidic dye is one in which the colour bearing ion, the chromophore, is an anion.
- A basic dye is one in which the colour bearing ion, the chromophore, is a cation.
- A neutral dye is a complex salt of a dye acid with a dye base.

Acid dyes generally combine more strongly with cytoplasmic (basic) elements of the cell, and basic dyes combine best with nucleic acid (acidic) elements of the cell. Table 3.2 shows the chemical characteristics of a stain or dye.

3.4 Principle of Staining

Positive Staining

In positive staining, the surface of the bacterial cell takes on the colour of the stain. When basic stain is applied, there is an attraction between the negatively charged cell surface and positively charged chromophore, which leads to staining of the cell (Figure 3.2).

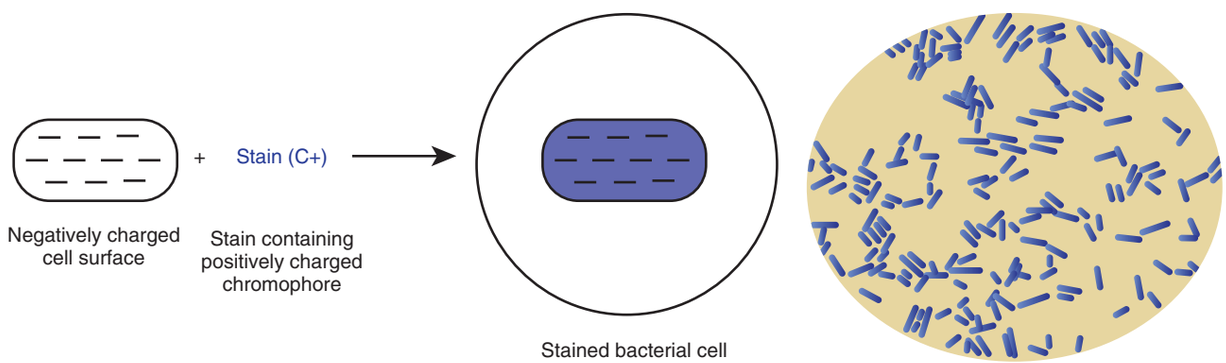


Figure 3.2: Positive staining

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